INCH-POUND

MIL-DTL-9320D
3 November 2008
SUPERSEDING
MIL-S-9320C
29 April 1971

DETAIL SPECIFICATION

SIGNAL, WARNING, AUDIBLE, FOR HEADSET, TYPE MA-1

Inactive for new design after 28 September 1999.

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of audible warning signal designated MA-1 for operation on 28 volts (V) direct current (DC).

1.2 International standardization agreement number.

STANAG 3370 (see 6.4)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:07Z
Check the source to verify that this is the current version before use.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

INTERNATIONAL STANDARDIZATION AGREEMENTS

STANAG 3370	- Aircrew Station Alerting Systems.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MS3106	- Connector, Plug, Electric, Straight, Solder Contacts, AN Type.
MIL-S-7742	- Screw Threads, Standard, Optimum Selected Series: General Specification for.
MIL-DTL-31000	- Technical Data Packages.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-704	- Aircraft Electric Power Characteristics.
MIL-STD-810	- Environmental Engineering Considerations and Laboratory Tests.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1916	- DoD Preferred Methods for Acceptance of Product.

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-781	- Handbook for Reliability Test Methods, Plans, and Environments for Engineering, Development Qualification, and Production.
MIL-HDBK-831	- Preparation of Test Reports.

(Copies of these documents are available at http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:07Z
Check the source to verify that this is the current version before use.

2.2.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

STANDARDIZATION DOCUMENTS

SD-6 - Provisions Governing Qualification.

(Copies of these documents are available at http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Warning signals furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.3).

3.2 Materials. Materials shall conform to the applicable specifications and shall be as specified herein. Materials that are not covered by specification or are not specifically described herein shall be of the best quality, of the lightest practicable weight, and shall be suitable for the purpose intended.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

3.2.2 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.2.3 Metals. Metals shall be corrosion resistant or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions that may be encountered in storage or normal service.

3.2.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other.

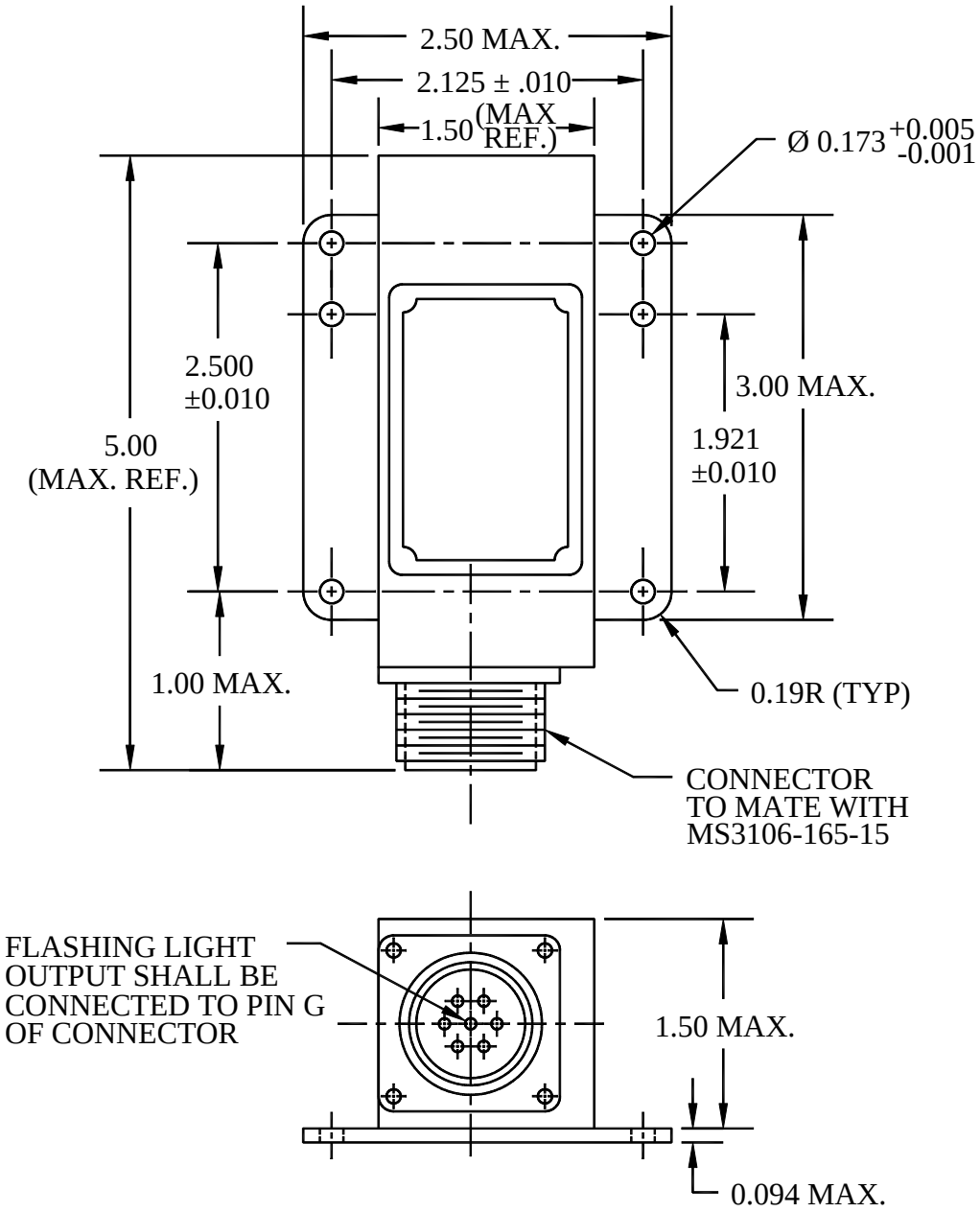
3.3 Design and construction. The warning signal shall be hermetically sealed, compact, and sufficiently rugged to the withstand shock and stresses incident to use on aircraft. Due to the high operating temperature requirement, the use of germanium diodes or transistors shall not be

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:07Z
Check the source to verify that this is the current version before use.

permitted. The design shall be completely composed of solid-state circuitry and shall not contain moving parts.

3.3.1 Adjustment. The warning signal shall be permanently adjusted at the time of manufacture and shall not require further adjustments.

3.3.2 Connections. Connections to the warning signal shall be made as shown on figure 1.



Dimensions in inches.
Decimals 0.XX ± 0.03.

FIGURE 1. Warning signal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:07Z
Check the source to verify that this is the current version before use.

3.3.2.1 Safety wiring. Means shall be provided for body attachment of safety wiring.

3.3.3 Input transient voltage protection. Components of the warning signal shall be protected against high-voltage spikes of 200 V DC of 1 microsecond time duration and against transients as specified in MIL-STD-704.

3.4 Performance.

3.4.1 Frequency and interruption rate of the warning signal. When tested as specified in 4.5.7, the frequency and interruption rates shall be:

a. Fundamental audio output frequency 250 ± 50 Hertz (Hz).
b. Interruption rate 5 ± 1 Hz.

3.4.1.1 Output audio cycle. When tested as specified in 4.5.6 and interrupted at the specified rate, the on-off cycle of the audio output shall be 50 percent on, 50 percent off, with a 5 percent tolerance in the on/off cycle.

3.4.1.2 Harmonic content. When tested as specified in 4.5.16 and energized with 28 V DC and loaded as specified in 3.4.2.2, the warning signal shall have an output signal containing both odd and even harmonics. The voltage readings of harmonics 2 through 11 shall total 120 percent to 220 percent of the fundamental voltage reading. The voltage readings of the odd harmonics shall total a minimum of 75 percent of the total harmonic count, excluding the fundamental.

3.4.1.3 Audio and flashing lamp output circuits. When tested as specified in 4.5.6, the interrupted audio output and flashing lap output shall be as shown in table I. The power measured shall be true to root-mean square (RMS) power. The 0 decibel (dB) shall be the base of 1 milliwatt (mW) into 150 ohms.

TABLE I. Audio and flashing lamp output.

Connector pin B	DB level	Flashing rate
C- and ground	22 ± 2	--
D- and ground	16 ± 3	--
E- and ground	9 ± 3	--
G- and ground	Flashing lamp output	5 ± 1 Hz

3.4.1.4 Steady tone. When tested as specified in 4.5.6, a steady tone of the specified frequency, uninterrupted, shall be provided between connector terminals F and ground when the signal is energized. The output of this tone shall be 19 ± 3 dB and the 0 dB shall be the base of 1 mW.

3.4.1.5 Flashing lamp circuit. When tested as specified in 4.5.6 and interrupted at the specified rate, the on-off cycle of the flashing light shall be 50 percent on, 50 percent off, with a 5 percent tolerance in the on/off cycle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:07Z
Check the source to verify that this is the current version before use.

3.4.2 Electrical circuits.

3.4.2.1 Power input. Terminal A of the connector shall be connected to 28 V DC positive. The unit shall be energized by connection of terminal B through external circuitry to 28 V DC negative.

3.4.2.2 System loading. The warning signal shall be compatible with the aircraft communication system. The portion of the communication system into which the warning signal shall operate is essentially an audio transformer terminated in a 9 ohms resistive headset. The transformer should have the following characteristics: Line type, primary impedance 150 Ohms, primary DC resistance of 7.5 ohms maximum, a secondary impedance of 9 ohms, a secondary DC resistance of 4.5 ohms maximum, a frequency response of 200 Hz to 6,000 Hz, and an output of 200 ± 5 mW.

3.4.2.3 Electrical output characteristics. When not energized, the warning signal shall present an open circuit to associated communications equipment. The resistance between any output circuit and terminal A of the connector shall be not less than 10 megohms. When not energized and 28 V DC is applied to pin B and pin C is grounded, the continuous current flow is to be less than 1.0 milliampere (mA). When the signal warning is energized, it shall be impossible for uninterrupted direct current to be conducted into the associated communication equipment.

3.4.2.4 Power output requirement. When operating on 28 V DC, the warning signal shall be capable of delivering 25 mA average direct current, minimum, into a 150-ohm load.

3.4.3 Dielectric strength. When tested as specified in 4.5.2, the warning signal shall withstand a voltage of 600 V root mean square (RMS) for 60 seconds when applied simultaneously between all connector terminals and the case.

3.4.4 Current consumption. When tested as specified in 4.5.3, the average current consumed by the warning signal shall not exceed 0.20 amperes when operating on 29 V DC.

3.4.5 Operating voltage. When tested as specified in 4.5.4, the warning signal shall operate as specified in 3.4.1 from 27 V DC to 29 V DC. The warning signal shall operate without incurring damage between 14 V DC to 29 V DC.

3.4.6 Environmental conditions. The warning signal shall operate electrically under the following conditions:

a. Vibration: When tested as specified in 4.5.9, the warning signal shall withstand typical aircraft vibration. Vibration isolators shall not be used.
b. Shock: When tested as specified in 4.5.10, the warning signal shall withstand a shock of 50 g.
c. High altitude - low temperature: When tested as specified in 4.5.11, the warning signal shall withstand pressure equivalent to 80,000 feet altitude and temperature of -65 °C.
d. Temperature: When tested as specified in 4.5.13 and 4.5.14, the warning signal shall operate without failure from -65 °C to 125 °C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:07Z
Check the source to verify that this is the current version before use.

3.4.7 Hermetic seal. When tested as specified in 4.5.12, the warning signal shall be hermetically sealed to withstand a pressure of 20 millimeters of mercury (mmHg) while submerged in water or ethylene glycol.

3.4.8 Short circuit. When tested as specified in 4.5.15, the warning signal and audio transformer shall withstand a short circuit for a minimum of 30 seconds from any output pin to ground.

3.4.9 Pressurization. When tested as specified in 4.5.17, the warning signal shall withstand pressure of 28 pounds per square inch (psi).

3.4.10 Reliability. When tested as specified in 4.5.8, the warning signal shall exhibit a minimum acceptable mean time between failure (MTBF) of 500 hours at 90 percent confidence level.

3.5 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-DTL-31000 shall govern the manufacturer's part numbers and changes thereto.

3.6 Dimensions. The warning signal shall conform to the dimensions shown on figure 1.

3.7 Weight. The weight of the warning signal shall be not more than 6 ounces.

3.8 Screw threads. Machine screw threads shall conform to MIL-S-7742.

3.9 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.10 Workmanship. The warning signal, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, accuracy of dimensions, and marking of parts and assemblies.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Test conditions. Unless otherwise specified, all tests shall be made at an atmospheric pressure of approximately 29.92 inches Hg and a temperature of 25 °C ± 5 °C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:07Z
Check the source to verify that this is the current version before use.

4.3 Qualification inspection.

4.3.1 Test samples. The qualification test samples shall consist of five signals representative of the production equipment. The samples shall be identified with the manufacturer's part number and any additional information required by the procuring activity.

4.3.2 Test report. After completion of the qualification tests, the contractor shall prepare a test report summarizing the test results. The contractor may refer to MIL-HDBK-831 for guidance in preparing the test report.

4.3.3 Qualification tests. Qualification tests shall consist of all the tests specified in 4.5. At the conclusion of each environmental test, the signal shall be tested for operation and audio output. The tests shall be performed on the samples as indicated and in the order listed in table II.

TABLE II. Qualification test schedule.

Test	Sample number				
	1	2	3	4	5
Individual tests (see 4.4.1)	X	X	X	X	X
Internal impedance (see 4.5.5)	X	X	X	X	X
Audio and flashing lamp output circuits (see 4.5.6)	X	X	X	X	X
Frequency and interruption rate (see 4.5.7)	X	X	X	X	X
Reliability (see 4.5.8)	X			X	
Vibration (see 4.5.9)		X			
Shock (see 4.5.10)					X
High altitude-low temperature (see 4.5.11)			X		
Pressurization (see 4.5.17)			X	X	
High temperature (see 4.5.13)					X
Low temperature (see 4.5.14)		X			
Short circuit (see 4.5.15)	X	X			
Harmonic content (see 4.5.16)	X	X	X	X	X
Hermetic seal (see 4.5.12)			X	X	

4.4 Conformance inspection. Conformance inspection shall consist of:

a. Individual tests (see 4.4.1).
b. Sampling tests (see 4.4.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:07Z
Check the source to verify that this is the current version before use.

4.4.1 Individual tests. Each signal shall be subjected to the following tests as described in 4.5:

a. Examination of product (see 4.5.1).
b. Dielectric strength (see 4.5.2).
c. Current consumption (see 4.5.3).
d. Operating voltage (see 4.5.4).

4.4.2 Sampling tests. Unless otherwise specified (see 6.2), sampling shall be in accordance with verification level I of MIL-STD-1916.

4.4.2.1 Lot. A lot shall consist of warning signals manufactured under essentially the same conditions and submitted for inspection at the same time.

4.4.2.2 Tests. The warning signal shall be subjected to:

a. Individual tests (see 4.5.1).
b. Internal impedance (see 4.5.5).
c. Reliability (see 4.5.8).
d. Vibration (limit test to room temperature) (see 4.5.9).
e. Shock (see 4.5.10).
f. High temperature (see 4.5.13).

4.5 Test methods.

4.5.1 Examination of product. The warning signal shall be inspected to determine compliance with the requirements specified herein with respect to material, design and construction, workmanship, dimensions, weight, and identification of product.

4.5.2 Dielectric strength. The warning signal shall be subjected to 600 V RMS for 60 seconds. The voltage shall be applied simultaneously between all terminals of the connector and the case. The warning signal dielectric strength shall meet the requirements specified in 3.4.3. At the conclusion of this test, there shall be no evidence of failure and the warning signal shall operate satisfactorily.

4.5.3 Current consumption. The output pins of the warning signal shall be connected to a 150-ohm load and the current measured. The warning signal current shall be as specified in 3.4.4. This test shall be performed separately on each output pin.

4.5.4 Operating voltage. The warning signal shall be monitored with an oscilloscope to determine its operation from a 14 to 29 V DC source. Frequency, dB level, and interruption rate shall be monitored when the warning signal is loaded as specified in 3.4.2.2. The warning signal operating voltage shall be as specified in 3.4.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:07Z
Check the source to verify that this is the current version before use.

4.5.5 Internal impedance. The impedance between pin A and each output pin shall be measured with pin A connected to 28 V DC positive, pin B disconnected, and an ammeter connected between the output terminal and the ground. A DC microammeter shall be used and the maximum current shall not exceed 2.8 microamperes. The current between pin B and pin C, with the unit not energized, shall be measured with 28 V DC applied to pin B and an ammeter connected between pin C and ground. This current shall be less than 1.0 mA.

4.5.6 Audio and flashing lamp output circuits. The dB levels shall be measured with a thermovoltmeter connected across a 150-ohm resistive load. The 150-ohm load shall be the combined parallel impedance of the thermovoltmeter used and the load resistor. The voltage readings shall be converted to dB readings. The 0 dB reading shall be the base of 1 mW into the 150 ohm load. Direct current of 28 V DC shall be applied to the warning signal. Terminal G of the connector shall be connected to the flasher circuitry internally and to an MS-type lamp externally. The warning signal dB levels and flashing lamp on/off cycle shall be as specified in 3.4.1.1, 3.4.1.3, table I, 3.4.1.4, and 3.4.1.5.

4.5.7 Frequency and interruption rate. When loaded as specified in 3.4.2.2 and tested by monitoring the output warning signal with an oscilloscope, the warning signal shall meet the requirements of 3.4.1 and 3.4.1.1.

4.5.8 Reliability.

4.5.8.1 Reliability test plan and report. Prior to starting the reliability test, the manufacturer shall submit a detailed reliability test plan for formal approval by the procuring activity. The manufacturer may refer to test plan I-D of MIL-HDBK-781 as a guide for the preparation of the reliability test plan. The manufacturer shall prepare a final report that summarizes the achieved reliability of the test samples. The report shall also contain test procedures, details of any malfunctions, and applicable charts and graphs showing failures/MTBF plotted against test time and other pertinent test data. The manufacturer may refer to the guidelines of MIL-HDBK-831 for preparation of the test report.

4.5.8.2 Reliability test. The warning signal shall be tested for reliability with a duty cycle of 250 hours continuous operation, followed by 250 hours cycling, 1 minute on and 1 minute off, at a normal input voltage and simulated load of 150 ohms between pin C and ground. Following the test, the warning signal shall meet the output requirements of 3.4.1.1.

4.5.9 Vibration. The warning signal shall be subjected to a vibration test in accordance with MIL-STD-810 Method 514, "Vibration". The warning signal shall meet the vibration requirements specified in 3.4.6 (a).

4.5.10 Shock. The warning signal shall be subjected to a shock test in accordance with MIL-STD-810 Method 516 "Shock", except that the magnitude of each shock impulse shall be 50 g. The warning signal shall meet the shock requirements specified in 3.4.6 (b).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:07Z
Check the source to verify that this is the current version before use.

4.5.11 High altitude - low temperature. The warning signal shall be placed in an altitude chamber maintained at a temperature of -65 °C and the pressure reduced to simulate an altitude condition of 80,000 feet for 4 hours. The warning signal shall meet the high altitude - low temperature requirements of 3.4.6 (c). After returning to room temperature, the warning signal shall be examined for any evidence of damage and shall operate as specified herein.

4.5.12 Hermetic seal. The warning signal shall be placed in a transparent container filled with ethylene glycol or water and placed in an altitude chamber equipped with a view plate or bell jar. The pressure shall be reduced to 20 mmHg or less (approximately 81,000 feet altitude) and maintained at this pressure for 30 minutes. Any leakage as indicated by bubbles shall constitute failure.

4.5.13 High temperature. The warning signal shall be subjected to a high temperature test in accordance with MIL-STD-810 Method 501 "High Temperature", except that the temperature shall be 125 °C. The warning signal shall meet the high temperature operation requirements of 3.4.6 (d). During the final hour of this test, the signal shall operate as specified herein.

4.5.14 Low temperature. The warning signal shall be subjected to a low temperature test in accordance with MIL-STD-810 Method 502, "Low Temperature", except that the temperature for the entire test shall be -65 °C. The warning signal shall meet the low temperature operation requirements of 3.4.6 (d). During the final hour of the test, the warning signal shall be energized and shall operate as specified in 3.4.

4.5.15 Short circuit. All output pins of the warning signal shall be individually connected to ground and the warning signal energized for not less than 30 seconds. The warning signal shall withstand a short circuit in accordance with the requirements specified in 3.4.8. Upon completion of this test, the warning signal shall operate as specified herein.

4.5.16 Harmonic content. Measurements shall be made on all four output pins separately using a harmonic analyzer connected across the secondary of the audio transformer when loaded as specified in 3.4.2.2. The measurements shall meet the requirements specified in 3.4.1.2.

4.5.17 Pressurization test. The warning signal shall be placed in a chamber with absolute pressure of 28 psi and a temperature of 25 °C for 4 hours. During this period, no deformation or other damage to the warning signal shall occur. At the conclusion of this test, the warning signal shall be closely examined for evidence of deformation and shall operate as specified herein. Evidence of deformation, damage, or any unsatisfactory operation shall be cause for rejection.

4.6 Maintenance of qualification. At specified intervals determined by the qualifying activity, the manufacturer must be able to demonstrate that the company still has the capabilities and facilities necessary to produce the QPL item(s) in accordance with this specification and with the provisions governing qualification in accordance with SD-6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:07Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the inventory control point's packaging activities within the military service or defense agency, or within the military service's system command. Packaging data retrieval is available from the managing military department's or defense agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The MA-1 warning signal is for use on aircraft to warn the pilot of an unsafe landing gear condition through injection of an interrupted tone in his headset and a flashing lamp indicator in the cockpit.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The specific issue of individual documents referenced (see 2.2.1 and 2.2.2).
c. Sampling, if different (see 4.4.2).
d. Packaging (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products that are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-9320 whether or not such products have actually been listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the federal government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Richmond, Attn: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297 or STDZNMGT@dla.mil.

6.4 International standardization agreement implementation. This specification implements STANAG 3370, "Aircrew Station Alerting Systems". When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standard agreement, as identified in the ASSIST database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:07Z
Check the source to verify that this is the current version before use.

6.5 Subject term (key word) listing.

aircraft
flashing
indicators
interrupted tone
lamp
landing gear

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AV
Navy - AS
Air Force - 99

Preparing Activity:
DLA - GS2

(Project 6340-2007-001)

Review Activities:
Air Force - 11, 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at http://assist.daps.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:07Z
Check the source to verify that this is the current version before use.